The simplex as the unique higher-rank projective general linear polytope

Abstract

We apply the common projective-line reduction to all regular polytopes of rank at least four with full automorphism group $\mathrm{PGL}_2(q)$. The unique example is the 4-simplex, with $\mathrm{PGL}_2(5) \cong S_5$. The proof includes all possible quotients of the finite universal groups and uniqueness of the distinguished marking. This is the classification of Leemans and Schulte, obtained here as a second complete application of the same centralizer and finite-amalgam theory.

1. Statement and shared hypotheses

We use the marked string C-group convention of [E], and the standard correspondence with regular abstract polytopes [MS]. In particular an isomorphism must preserve the ordered distinguished generators, not merely the abstract automorphism group. A regular polytope's full automorphism group acts regularly on its flags.

Theorem 1 (Leemans–Schulte). Let q be a prime power. If a regular abstract polytope P of rank $n \geq 4$ has full automorphism group isomorphic to $\mathrm{PGL}_2(q)$, then $n = 4$, $q = 5$, and P is the 4-simplex. Its distinguished marking is the standard adjacent-transposition marking of S_5, of type $\{3, 3, 3\}$. Conversely that marking gives the required polytope.

By [E, Theorem 1], the hypotheses already force odd q and rank four. Both end parabolics are proper and have groups S_4 or A_5. All their ordered rank-three markings and compatible pairs are included in [E, Lemmas 10–11]. Consequently the desired group is a marked quotient of one of ten explicit universal groups. The common reduction includes the cases of subfield PSL_2 groups, subfield PGL_2 groups, affine groups and unipotent dihedral subgroups; none has been omitted as a small-field exception.

2. Elimination of every other quotient

Four universal groups in the table are trivial. In the rows of orders 60 and 24, an end parabolic equals the entire universal group. Passing to a quotient preserves this generator dependence, contradicting the marked intersection property. These six rows cannot produce a polytope in the statement.

The two universal groups $\mathrm{PSL}_2(11)$ and $\mathrm{PSL}_2(19)$ are nonabelian simple, so their only nontrivial quotients are themselves. For odd q the determinant modulo squares gives $\mathrm{PGL}_2(q)$ a quotient of order two. It cannot be either of those simple groups. This eliminates the two locally projective rows without comparing orders or silently replacing $\mathrm{PGL}_2(q)$ by $\mathrm{PSL}_2(q)$.

The universal group in the remaining row of order 96 can map onto $\mathrm{PGL}_2(q)$ only if $q(q^2 - 1)$ divides 96. For odd $q \geq 5$ the order is at least 120; thus the only possibility is $q = 3$. But $\mathrm{PGL}_2(3) \cong S_4$ cannot contain the required S_4 or A_5 facet as a proper subgroup. Hence that row also gives nothing.

The final universal group is S_5, of order 120. Its normal subgroups are 1, A_5, S_5, so the only nontrivial quotient groups are S_5 and C_2. The order of $\mathrm{PGL}_2(q)$ for odd $q \geq 3$ is at least 24, excluding C_2. Its order $q(q^2 - 1)$ is strictly increasing in positive q, and equals 120 precisely at $q = 5$. The remaining group is therefore $\mathrm{PGL}_2(5) \cong S_5$. This proves exhaustion for every prime power, including the small cases excluded by the common rank theorem.

3. The marking, existence and uniqueness

The successful local pair has both facet and vertex figure the tetrahedron, with presentation $[3, 3]_4$ in [E]. Its four-generator universal presentation is

$$\begin{aligned}\langle r_0, r_1, r_2, r_3 \mid r_i^2 = 1,\ (r_i r_j)^2 = 1 \quad (j - i > 1),\\ (r_0 r_1)^3 = (r_1 r_2)^3 = (r_2 r_3)^3 = 1,\\ (r_0 r_1 r_2)^4 = (r_1 r_2 r_3)^4 = 1\rangle.\end{aligned} \tag{2}$$

The shared exact computation proves that this is S_5 with its specified string C-marking. There is also an explicit geometric identification. Send the generators in order to (12),(23),(34),(45). These adjacent transpositions generate S_5; they satisfy all of (2), including the two 4-cycle relations. The resulting epimorphism from the universal group is an isomorphism because both groups have order 120. Thus its marking is precisely the familiar simplex marking, not merely a group of the same order as the simplex. By [MS] this gives the 4-simplex, and hence existence in Theorem 1.

Every possible local marking is covered by the common six-class enumeration. For each compatible pair, its literal universal presentation supplies the marked epimorphism to the group under consideration. Section 2 eliminates all pairs except the tetrahedral pair, and eliminates every proper nontrivial quotient of its universal group. Any surviving epimorphism is therefore an isomorphism preserving the four distinguished generators. This proves uniqueness of the marked group and of the polytope. Reversing the four adjacent transpositions is also an automorphism, by conjugation with the permutation reversing 1, 2, 3, 4, 5, so the usual self-duality is recovered. Theorem 1 follows. ■

4. Source scope, attribution and proof dependencies

This is the complete principal contract of [LS, Theorem 10], together with its rank restriction and introductory rank-at-least-four formulation. The classification is attributed to Leemans and Schulte. The projective-line centralizer descent and the odd-order normalizer obstruction are supplied by the common paper [E]; its single finite program supplies all six local classes and ten universal amalgams. The PSL_2 application is a separate consequence of the same common paper, not an additional unproved premise of the argument here.

Dickson's subgroup types and their standard embeddings, elementary projective group structure and the marked group/polytope correspondence remain retained classical inputs. The shared finite-presentation calculations use GAP's exact algorithms. Its Lean component proves only the abstract normalizer obstruction; there is no end-to-end Lean proof of this application at this stage. No historical priority or new-conjecture claim is made.

The final part of [LS] also prints a seventeen-case census for $P\Sigma L_2(25)$, recalls other almost-simple and chiral constructions, and states Conjecture 11 about extensions at odd square q. Those are separate assertions. They are neither premises nor conclusions here, and their current status is not inferred from the projective general linear classification. The separate historical-status audit records the later answer to Conjecture 11; this application does not claim that answer. In particular the present paper is not a replacement for all statements in that source.

References

[E] *Centralizers, facets and finite amalgams on the projective line*, companion umbrella paper, especially Theorem 1 and Lemmas 10–11.

[LS] D. Leemans and E. Schulte, *Polytopes with groups of type* $\mathrm{PGL}_2(q)$, retained arXiv:0909.1991v1, 9 physical pages, Theorem 10 on page 7.

[MS] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, especially the string C-group correspondence.